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COMET FWC 070

Fourth International Olympiad, 1962

1962/1

Find the smallest natural number n which has the following properties:

1. Its decimal representation has 6 as the last digit.
2. If the last digit 6 is erased and placed in front of the remaining digits, the resulting number is four times as large as the original number n.

1962/2

Determine all real numbers x which satisfy the inequality:

$$\sqrt{3-x}-\sqrt{x+1}>\frac{1}{2}$$

1962/3

Consider the cube $ABCDA'B'C'D'$ (where $ABCD$ and $A'B'C'D'$ are the lower and upper bases, respectively, and edges AA', BB', CC', DD' are parallel).

The point X moves at constant speed along the perimeter of the square $ABCD$ in the direction $ABCDA$, and the point Y moves at the same rate along the perimeter of the square $B'C'CB$ in the direction $B'C'CB'$.

Points X and Y start from A and B', respectively.

Determine and draw the locus of the midpoints of the segments XY.

1962/4

Solve the equation

$$\cos^2 x+\cos^2 2x+\cos^2 3x=1$$

1962/5

On the circle K, there are given three distinct points A, B, C.

Construct (using straightedge and compass) a fourth point D on K such that a circle can be inscribed in the quadrilateral thus obtained.

1962/6

Consider an isosceles triangle.

Let r be the radius of its circumscribed circle and ρ the radius of its inscribed circle.

Prove that the distance d between the centers of these two circles is

$$d = \sqrt{r(r - 2\rho)}$$

1962/7

The tetrahedron $SABC$ has the following property:

There exist five spheres, each tangent to the edges SA, SB, SC, BC, CA, AB, or to their extensions.

1. Prove that the tetrahedron $SABC$ is regular.

2. Prove conversely that for every regular tetrahedron, five such spheres exist.